Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad — July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering — *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi — Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering — *CGPA: 9.05*

Research & Open-Source Experience

Core Contributor, RookDB — Aug 2026 – Present
Independent Study, DSAC Lab, IIIT Hyderabad | Rust, Query Optimization, Benchmarking — *Hyderabad, India*

- Co-architected the lab's open-source SQL-99 RDBMS in Rust across storage, parser and CLI crates, with **1,200+ tests**.
- Built column statistics and a cost model choosing hash vs. index joins and join order, checked against `EXPLAIN ANALYZE`.
- Differential testing matches SQLite on **50,000 random queries**; stress benchmarks cut the 1M-row suite **14x (827 s to 58 s)**.

Projects

Peer-to-Peer Distributed File Sharing System | *C++, Multithreading, Sockets, OpenSSL, Linux*

- Built a BitTorrent-style P2P system in C++; self-scheduling worker threads and a fix for a Nagle/delayed-ACK stall in length-prefixed TCP framing cut 1GB single-peer downloads from **193s to under 13s**.
- Removed a data race and a file-wide lock by confining shared state to one mutex each and writing out-of-order chunks with lock-free positional `pwrite`, after a SHA-1 check of every 512KB piece.
- Architected Primary-Backup tracker failover with race-free election and state sync; **46 unit tests** pass under ASan/UBSan.

IndicLLM: Transformer Pretraining & Reasoning SFT | *Python, PyTorch, SentencePiece, ONNX*

- Pretrained four 25M-param Transformers from PyTorch primitives on **500M tokens each** on GPUs, verifying causality with **bit-identical** logits under future-token changes; RoPE/SwiGLU/RMSNorm cut test perplexity **5–7%**.
- Ran **44** seeded fine-tuning runs on solver-generated reasoning, reaching **0.96** test accuracy and **0.85** on 10-hop chains (3× the training depth) against a 0.48 majority baseline.
- Showed with a leakage analysis that a training-answer lookup scores 0.95 on the main test but chance on deeper splits, where the models still score **0.82–0.99**; found lower perplexity did not predict better reasoning.

Warehouse Traffic Control: Deterministic Distributed Simulator | *JavaScript, Web Workers, Three.js*

- Built a seed-replayable simulator of a 200-robot warehouse whose components talk only over a lossy, delayed, fault-injected network; checkpoint and action-log replay give **sub-0.2 s time-travel seeks**, verified by state-equivalence tests.
- Designed a lease-and-fencing protocol across 16 region managers, sustaining **0 collisions over 416K moves** under 40% message loss, ±20% clock drift and injected crashes.
- Ran parameter sweeps on a Web Worker pool (a 56-run comparison in ∼7 min on 8 cores); 30-minute runs **overturned two conclusions** drawn from 2-minute runs, including an apparent safety-margin sweet spot.

Technical Skills

Languages: C/C++, Python, Rust, SQL, Bash, JavaScript.

ML & Evaluation: Transformers, Pretraining & Fine-Tuning, Seeded Experiments, Ablations, Leakage Analysis.

Libraries: PyTorch, SentencePiece, ONNX Runtime, NumPy, Pandas, Scikit-learn.

Systems & Tools: Multithreading, Sockets, Profiling & Benchmarking, Linux, Git, Docker, ASan/UBSan.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.